

Communicating Uncertainty – Expectations and Facts

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What are the chances that...

- ...you will win the Eurojackpot main prize?
- ...China will attack Taiwan?
- ...Kishinev is the capital of Georgia?
- ...you will be hit by a car tomorrow?
- ...you could be a millionaire, but you missed the chance once?

What kind of information about chances would you like to obtain when deciding about...

- ...investing in the specific stock (e.g., Microsoft)?
- ...booking a bus ride, that should be one hour after the scheduled arrival of your plane?
- ...undergoing an operation in the hospital, that can completely eliminate your health issues, but it can also worsen them?

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My search for uncertainty

What to study?

Kahneman, Tversky 2002 Nobel Prize

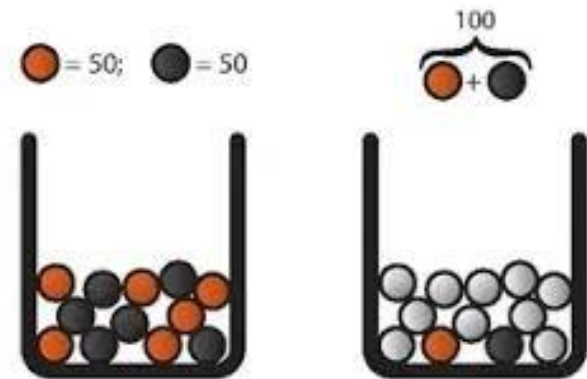
- decision-making in general – individual differences
- Decision-making styles (habitual patterns – rational, intuitive...)
- Decision-making competence
- Indecisiveness
- Intolerance of uncertainty

Uncertainty, risk, ambiguity

Two urns are filled with red and black balls. Urn A contains 50 red balls and 50 black balls randomly mixed; Urn B contains 100 red and black balls randomly mixed in an unknown ratio. Right bet = 10 EUR

You are drawing from the first urn.
Which colour you bet on?
And the second urn?

You want to draw the red ball.
Whic urn to choose?
And for the black one?



The same probabilities, but not the same perception – preferring known distributions

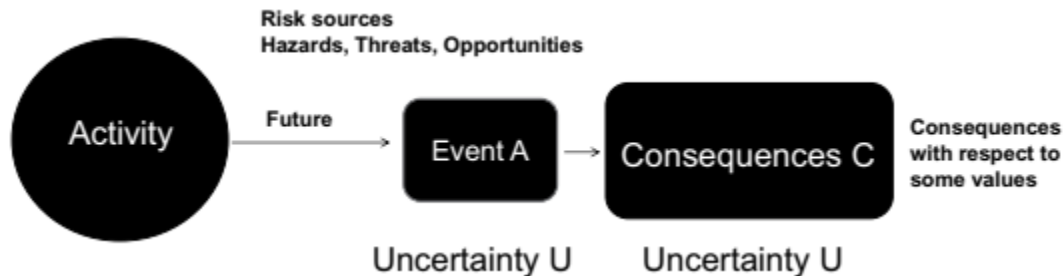
Risk as outcomes with known probabilities – as an antipole of **ambiguity** (present state) or **uncertainty** (future state)

Uncertainty, risk, ambiguity

Risk

- a) the potential for undesirable consequences
- b) the consequences of the activity and associated uncertainties
- c) the deviation from a 'reference value', and associated uncertainties

Event is referred as a hazard, threat or opportunity – or a risk source (Aven & Thekdi, 2022)



Risk – a) future, negative or b) future, known probabilities

Uncertainty – a) not known or b) future, unknown probabilities

Ambiguity – present, not known (incomplete information)

Where is uncertainty?

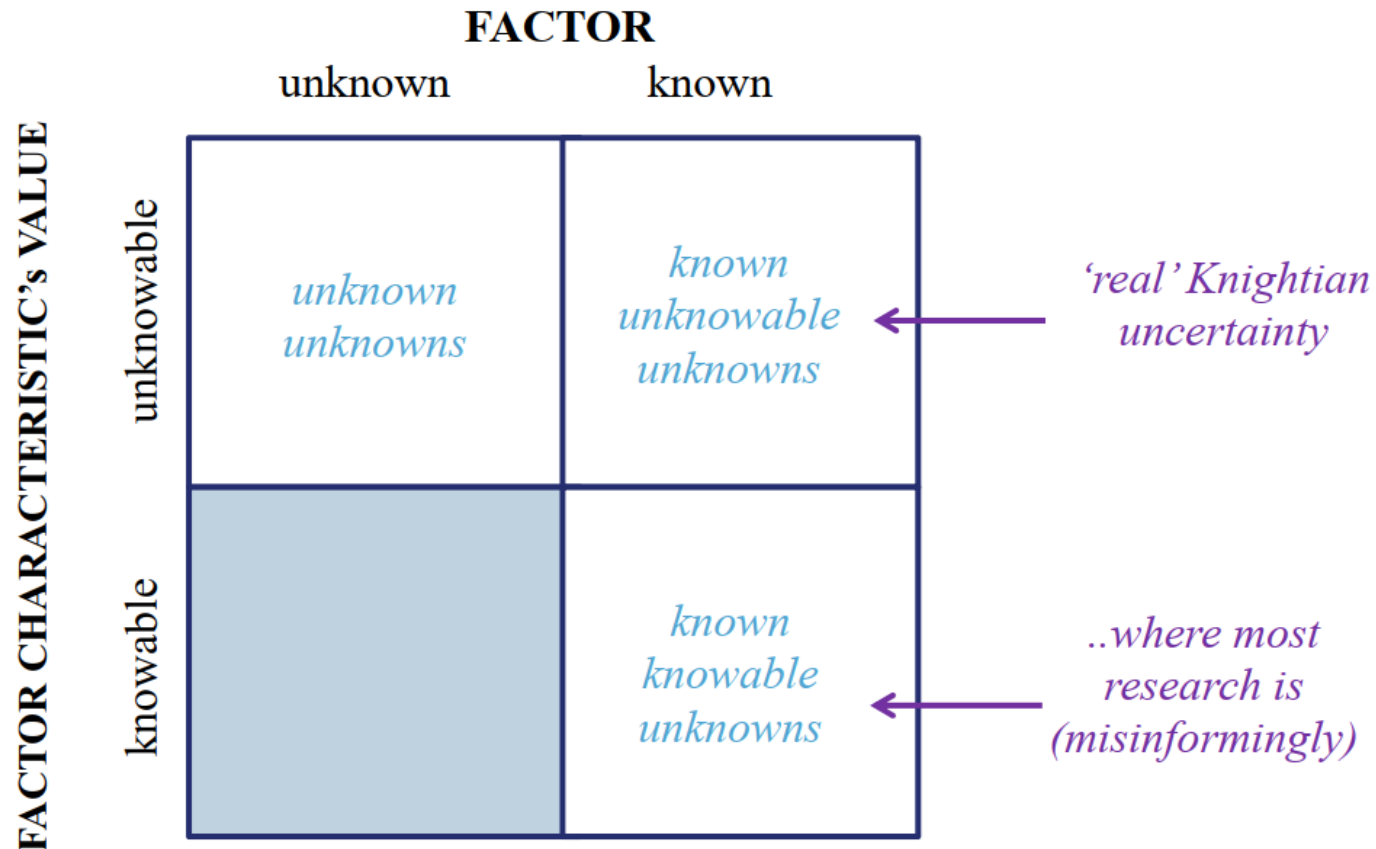
„In this world, nothing can be said to be certain, except death and taxes“ (Benjamin Franklin)

I have approximate answers and possible beliefs in different degrees of certainty about different things, but I'm not absolutely sure of anything. (Richard Feynman)

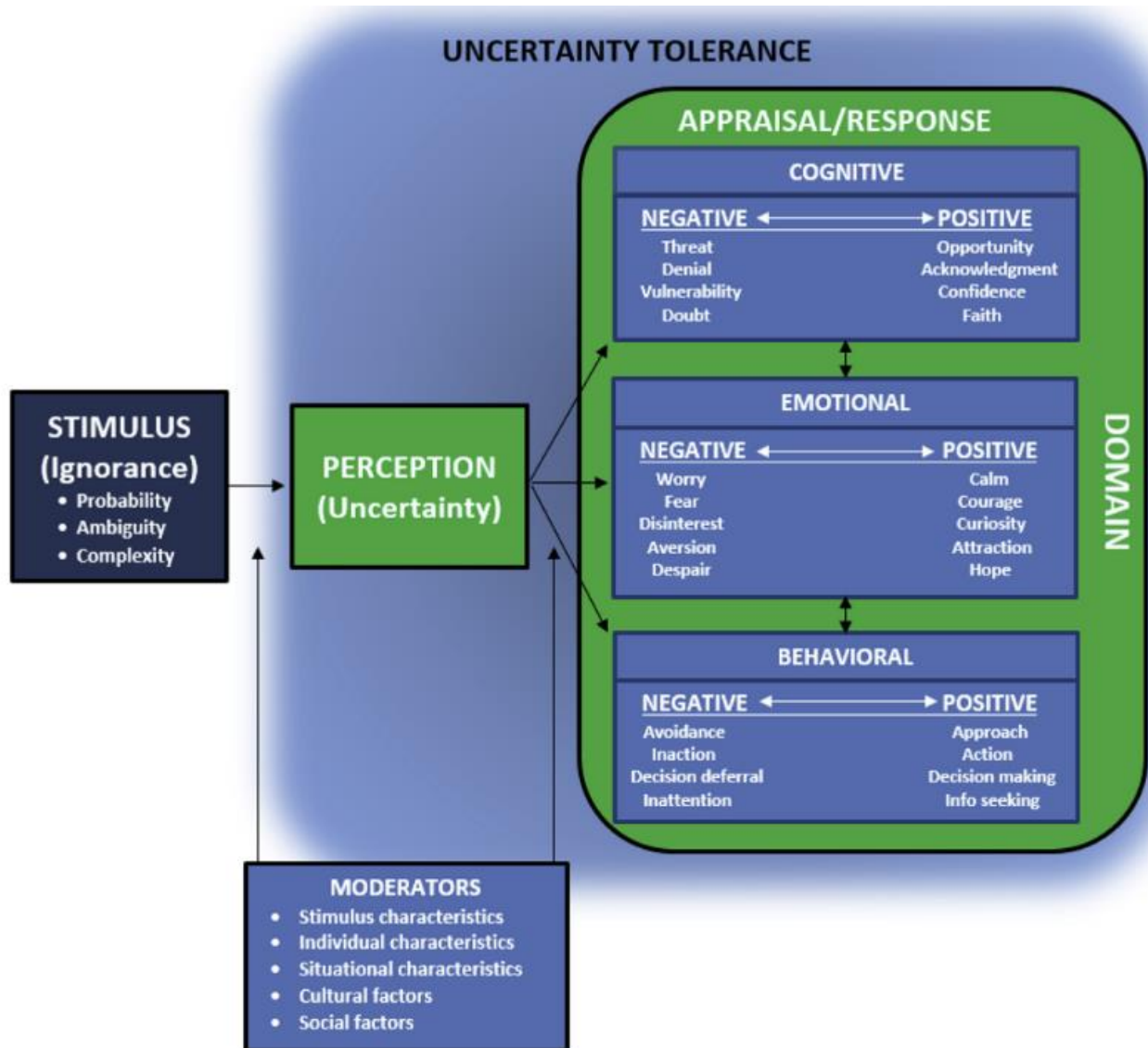
„Reports that say that something hasn't happened are always interesting to me, because as we know, there are known knowns; there are things we know we know. We also know there are known unknowns; that is to say we know there are some things we do not know. But there are also unknown unknowns—the ones we don't know we don't know.“
(Donald Rumsfeld, US Secretary of Defense, 2002 - evidence linking the government of Iraq with the supply of weapons of mass destruction to terrorist groups)

Types of uncertainty

- a) Epistemic uncertainty – subjective – we do not have the information (when will the train arrive? – endogenous) – *I am not certain*
- b) Aleatory – objective – what will happen (will it rain next Saturday? – exogenous) – *it is not certain*



Reactions to uncertainty



- Individual differences
- Intolerance of uncertainty**
- Prospective anxiety – prepare for uncertainty
 - Inhibitory anxiety – try to avoid uncertainty
- Intolerance of ambiguity** (attitude to ambiguity)
- usually negative correlates

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Expression of uncertainty in communication

- Remember your answers to the questions at the beginning

Two broad classes of uncertainty expression and communication

Numerical – percentages (80%), proportions (0.80), fractions (4/5), odds (4:1)

- More precise, preferred by recipients

Verbal – probable, possible, slight hope

- More common, preferred by communicators

Epistemic uncertainty are more likely to be conveyed using words, whereas distributional (aleatory) uncertainty is more likely to be expressed numerically

Graphical methods

Expression of uncertainty in communication

- One of the main questions – the association (correspondence) between two modes – can we translate the verbal expressions to numbers?

Table 2 Approximate probability scales recommended by EFSA (European Food Safety Authority), IPCC (Intergovernmental Panel on Climate Change), and NATO (North Atlantic Treaty Association).

Adapted from European Food Safety Authority et al. (2018, Table 4), and Mastrandrea et al. (2010), and Allied Joint Doctrine for Intelligence Procedures AJP-2.1 (in Irwin & Mandel, 2020)

EFSA (2018)		IPCC (2010)		NATO (2016)	
Probability term	Subjective probability range	Term	Likelihood of outcome	Verbal statement	Numerical assessments
Almost certain	99–100%	Virtually certain	99–100%		
Extremely likely	95–99%				
Very likely	90–95%	Very likely	90–100%	Highly likely	More than 90%
Likely	66–90%	Likely	66–100%	Likely	60–90%
About as likely as not	33–66%	About as likely as not	33–66%	Even chance	40–60%
Unlikely	10–33%	Unlikely	0–33%	Unlikely	10–40%
Very unlikely	5–10%	Very unlikely	0–10%	Highly unlikely	Less than 10%
Extremely unlikely	1–5%				
Almost impossible	0–1%	Exceptionally unlikely	0–1%		

Expression of uncertainty in communication

- One of the main questions – the association (correspondence) between two modes
– can we translate the verbal expressions to numbers?

Table 2 Form of words for drug adverse events mandated by the European Medicines Agency (EMA 2016) for information inserts in drug packaging

Verbal Description	Frequency group
Very common	$\geq 1/10$
Common	$\geq 1/100$ to $< 1/10$
Uncommon	$\geq 1/1,000$ to $< 1/100$
Rare	$\geq 1/10,000$ to $< 1/1,000$
Very rare	$< 1/10,000$
Frequency not known (cannot be estimated from the available data)	

Verbal and numerical expressions – specific outcome vs range

Table 1 Facets of meaning conveyed by verbal and numeric statements about uncertain outcomes

Verbal probabilities of specific outcomes			Uncertain outcomes as numeric ranges	
Section		Examples	Section	Examples
1	Degrees of probability	Numeric translations of verbal phrases “Likely” means 70%	1	Degrees of uncertainty Interval width 40–60 cm vs. 20–80 cm increase in sea level
2	Hedging	Politeness and understatements “It is possible”	2	Hedging Approximate estimates “About”, use of round numbers
3	Sources of probability	External vs. internal “it is uncertain” vs. “I am uncertain”	3	Sources of uncertainty External variability vs. imprecise knowledge
4	Expertise and credibility	Numbers suggest expertise The confidence heuristic	4	Expertise and credibility Preciseness suggests expertise (when plausible)
5	Valence	Positive vs. negative outcomes “A hope” vs. “a risk”	5	Comparative judgments Exclusive vs. inclusive ranges “More than–less than” vs. “at least–at most”
6	Directionality of VPE	Framing with verbal expressions “Likely” vs. “not completely certain”	6	Directionality of incomplete ranges (upper or lower bounds) Framing with single bounds “At least X” vs. “at most Y”
			7	Which of multiple or continuous outcomes correspond to a VPE The “Which Outcome” approach Extremity and centrality effects

Table source: Teigen, 2023

Expression of uncertainty in communication

- the size of an individual's verbal probability lexicon is, on average, no more than ten terms
- even the numeric interpretation of probability terms is rounded up or down to the nearest 0% or 5%
- words provide fewer distinguishable levels of probability, and receivers believe that terms do not provide enough information compared with point numeric values

Imagine a lottery with 1000 numbered tickets.

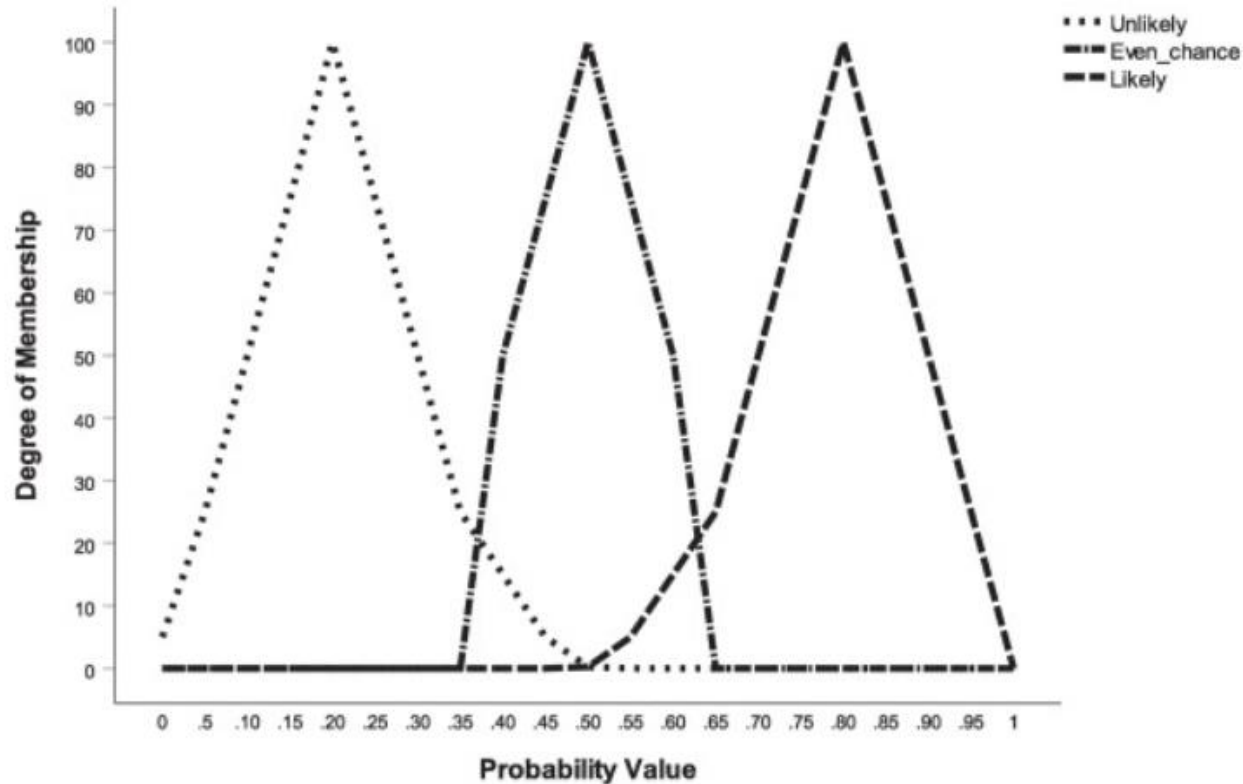
Ten prizes of 100 dollars each, and one top prize of 1000 dollars.

Complete the sentence:

- a) In this lottery, it is possible to win.....
- b) In this lottery, it is almost impossible to win.....

Expression of uncertainty in communication

- Inter-individual variability – membership function



Expression of uncertainty in communication

- Directionality of verbal expressions
 - Positive and negative expressions
 - 20% - slight hope vs minor concerns
- It is possible that the oil price will reach \$100 in March. 2/3
- It is not quite certain that the oil price will reach \$100 in March. 2/3
- Which forecaster has a higher probability in mind?
- Imagine that the oil price did in fact reach \$100. Who was more right?
- It is possible that oil prices will go up, because.....
- Implying action, providing recommendations – biasing decision?

Expression of uncertainty in communication

Teigen (and others) found:

1. A greater variety of positive (upward) than negative (downward) verbal probabilities, and they are used more often. The term “Likely” is much more frequent than the term “unlikely” in most texts.
2. VP statements may be vague in terms of probabilities, but they are not vague with respect to directionality (Teigen & Brun, 1995). People agree that “possible” is positive and “uncertain” is negative.
3. They act as frames, indicating higher or lower reference values (Juanchich et al., 2010; Honda & Yamagishi, 2017).
4. They are received by recipients as expressions of recommendations (or warnings), reflecting attitudes of speakers.
5. They influence decisions in opposite direction. A treatment will be tried when it is said to have some possibility of being helpful, but not when it is said to be somewhat uncertain (Teigen & Brun, 1999). Thus they are in a way more informative for decision makers than mere numbers (Collins & Mandel, 2019)
6. They facilitate or prevent conjunction fallacies. The conjunction of two “likely” events may be (mistakenly) perceived as more likely to occur than just one of them, while the combination of two “uncertain” outcomes is perceived as even more uncertain (Teigen & Brun, 1999).

Limitations of current approaches.....and challenges for the future

Driven by researchers – they provide the phrases

- What is your plan today?
- What is your work goal for the next week? How can it happen?
- identify expressions of uncertainty in linguistic data using probabilistic and large language models
- environmentally valid (natural) uncertainty expressions

Limitations of current approaches.....and challenges for the future

Focused on numbers – do we really have numbers in our mind when processing uncertainty?

- What is your plan today?
- What is your work goal for the next week? How can it happen?
- identify more common and neglected phrases like „can“, „may“
- describe (and quantify) their use

Limitations of current approaches.....and challenges for the future

English focused – neglecting other languages, missing multilingual studies

- mostly bilingual studies, comparing probability phrases in English and selected language
- provide a multilingual dictionary/lexicon of uncertainty phrases

Limitations of current approaches.....and challenges for the future

Neglecting inter-individual variability – most studies acknowledge it, but are satisfied with description

- what are the sources?
- identify sources of inter-individual variability – individual differences
- identify profiles according to attitudes to uncertainty (unknown)

Key messages

Uncertain as unknown – ambiguity, risk (vagueness, probability)

Usually negative outcomes and correlates

Affected by individual differences in attitudes to uncertainty

Two main modes of uncertainty expression – verbal and numerical

Preference affected by the position in communication

Directionality of verbal expressions

Inter-individual differences